

MATA37H3 — Week 14

Answer Key & Misconception Notes

For tutor use only — do not show the student until after their attempt. (Taylor Series Applications: Binomial Series, Error Bounds & Series Gymnastics)

EXAMPLE 1 — Answer

$\binom{-2}{0} = 1$, $\binom{-2}{1} = -2$, $\binom{-2}{2} = \frac{(-2)(-3)}{2} = 3$, $\binom{-2}{3} = \frac{(-2)(-3)(-4)}{6} = -4$. So $(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots = \sum (-1)^n (n+1)x^n$. Substituting $x \rightarrow -x$ gives $(1-x)^{-2} = \sum (n+1)x^n$ — matches Week 11 Problem 6 exactly.

Common misconception: Forgetting to alternate signs when r is a negative integer — the pattern $1, -2, 3, -4, \dots$ is easy to mis-copy as all-positive since the *magnitudes* $1, 2, 3, 4$ look like the familiar $(n+1)$ pattern on their own.

EXAMPLE 2 — Answer

$\binom{1/2}{0} = 1$, $\binom{1/2}{1} = \frac{1}{2}$, $\binom{1/2}{2} = \frac{(\frac{1}{2})(-\frac{1}{2})}{2} = -\frac{1}{8}$. So $\sqrt{1+x} \approx 1 + \frac{x}{2} - \frac{x^2}{8} + \dots$.

Common misconception: Assuming all binomial coefficients for $r = \frac{1}{2}$ stay positive (since $r > 0$) — the factors $(r-1), (r-2), \dots$ go negative as soon as $n \geq 2$, since $r-1 = -\frac{1}{2} < 0$ here.

EXAMPLE 3 — Answer

$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$, so $\cos x - 1 + \frac{x^2}{2} = \frac{x^4}{24} - \frac{x^6}{720} + \dots = x^4 \left(\frac{1}{24} - \frac{x^2}{720} + \dots \right)$. Dividing by x^4 and letting $x \rightarrow 0$: limit is $\frac{1}{24}$.

Common misconception: Stopping the $\cos x$ expansion at the x^2 term (matching only what's explicitly subtracted) and missing the x^4 term that produces the actual limit — same trap as Example 2.

EXAMPLE 4 — Answer

$\sin 1 \approx 1 - \frac{1}{6} + \frac{1}{120} = 1 - 0.16\bar{6} + 0.008\bar{3} \approx 0.84167$. Alternating Series Remainder: error $\leq$ next omitted term $= \frac{1}{5040} \approx 0.000198$, so $\sin(1) = 0.84167 \pm 0.0002$ (matches the true value $\sin 1 \approx 0.84147$).

Common misconception: Reporting the estimate without stating an error bound — the whole point of using the *Alternating Series* Remainder (rather than a vague “should be close”) is that it gives a **guaranteed, computable** bound on how far off the truncated sum can be.

EXAMPLE 5 — Answer

$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \frac{x^8}{24} - \dots$. Integrate term-by-term from 0 to 1:

$1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \dots \approx 1 - 0.3333 + 0.1 - 0.0238 + 0.00463 - \dots \approx 0.7475$. The 5th term (≈ 0.0046) already affects only the third decimal, so stopping after 5 terms gives ≈ 0.75 to two decimal places (matches the known value ≈ 0.7468 closely at this precision).

Common misconception: Integrating only the first two or three terms and declaring victory without checking that the omitted terms are actually small enough for the *requested* precision — always state which term you stopped at and why it's safely below the needed error.

EXAMPLE 6 — Answer

$p_2(0.3) = 1 - \frac{0.3^2}{2} = 1 - 0.045 = 0.955$. Since the series alternates in *sign* but the terms here are not simply “the next term in a numerical alternating series” in the same clean sense once truncated at a fixed order, use the Lagrange bound directly: $|R_2(0.3)| \leq \frac{M}{3!}(0.3)^3 = \frac{1}{6}(0.027) = 0.0045$. So $\cos(0.3) \approx 0.955 \pm 0.0045$ (matches the true value $\cos(0.3) \approx 0.9553$).

Common misconception: Trying to apply the Alternating Series Remainder here since $\cos x$'s *full* series alternates — but once truncated at a specific p_n , the correct tool for a guaranteed bound on the *function's* error (not a numerical series' error) is the Lagrange Remainder, which uses the next derivative's bound, not “the next term.”

EXAMPLE 7 — Answer

Convergence of the Taylor series (as a series, for whichever x make it converge) says nothing by itself about what it converges *to* — Theorem 6.8 says the series equals $f(x)$ **only when** $R_n(x) \rightarrow 0$ for that x . A series can converge everywhere yet its sum disagree with f off an isolated point (the standard example, not required here, is $f(x) = e^{-1/x^2}$ for $x \neq 0$, $f(0) = 0$: every derivative at 0 is 0, so its Maclaurin series is identically 0, which trivially “converges everywhere” — to 0, not to $f(x)$, for any $x \neq 0$). The extra ingredient is exactly the Lagrange remainder bound: you need $R_n(x) \rightarrow 0$, not just that the series of $\frac{f^{(n)}(a)}{n!}(x-a)^n$ happens to converge as a numerical series.

Common misconception: Treating “the Taylor series converges” and “the function is analytic” as synonyms — this is precisely the gap Theorem 6.8 exists to close, and it is a favorite trap on both problem sets and exams.